

# **Analysis of Cyberchondria, Metacognitive Beliefs and General Anxiety**

M.Sc. Dissertation Data Analysis

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**Software:** R, RStudio, MS Excel

**Dataset:** 229 participants

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## PROJECT OVERVIEW

### Objective

The objective of the project was to understand the relationship between cyberchondria, metacognitive beliefs and anxiety, with a particular focus on examining metacognitive beliefs as a potential mediator between cyberchondria and anxiety.

The sample size was estimated a priori using **G\*Power 3.1.9.7** and statistical analyses were conducted on **RStudio v4.5.3**. The project involved data preparation, descriptive statistics, normality testing, reliability analysis, correlation, regression and mediation analysis.

### Research Questions

- Does cyberchondria severity differ between men and women
- Do significant correlations exist between cyberchondria, metacognitive beliefs and anxiety?
- Which among metacognitive beliefs and anxiety is a stronger predictor of cyberchondria?
- Which dimensions of metacognitive beliefs contribute most to predicting cyberchondria?
- Do metacognitive beliefs explain the pathway through which anxiety influences cyberchondria?

## DATASET OVERVIEW

### Summary

- Dataset Type: Cross-section survey data
- Sample Size: 229
- Data Collection: Google Forms
- Data Preparation: Microsoft Excel
- Analysis Software: R using RStudio

### Variables

| Variable                                      | Scale Type                | Description                       |
|-----------------------------------------------|---------------------------|-----------------------------------|
| Gender                                        | Categorical               | Participant gender                |
| Cyberchondria Severity<br>(CSS – 12)          | Continuous (DV)           | Total Cyberchondria               |
| Generalized Anxiety<br>Disorder (GAD – 7)     | Continuous                | Total anxiety score               |
| Metacognitions<br>Questionnaire (MCQ –<br>30) | Continuous (IV/ Mediator) | Total metacognitive beliefs score |

### PROJECT WORKFLOW

| Procedure                                             | Software/ Tool  |
|-------------------------------------------------------|-----------------|
| Research Objective                                    |                 |
| Sample Size Estimation                                | G*Power         |
| Data Collection                                       | Google Forms    |
| Data Cleaning and Preparation                         | Microsoft Excel |
| Exploratory Data Analysis                             | RStudio         |
| Normality Testing ( <i>Shapiro-Wilk</i> )             |                 |
| Wilcoxon Rank sum Test                                |                 |
| Reliability Analysis ( <i>Cronbach's alpha</i> )      |                 |
| Correlation Analysis ( <i>Spearman's rho</i> )        |                 |
| Multiple Linear Regression and Regression Diagnostics |                 |
| Mediation (PROCESS Macro 4)                           |                 |

Data preparation involved the following steps:

- Removal of incomplete or invalid responses, where applicable.
- Calculation of total and subscale scores for the GAD-7, CSS-12, and MCQ-30 questionnaires according to their respective scoring guidelines.
- Organization of variables into a structured dataset suitable for statistical analysis.

Following data preparation, the cleaned dataset was imported into RStudio for statistical analysis.

## DATA ANALYSIS

**Table 1. Exploratory Data Analysis**

| Variable       | Mean  | Std. Dev | Min | Max | Range | Skew | Kurtosis |
|----------------|-------|----------|-----|-----|-------|------|----------|
| CSS – 12 Total | 29.71 | 9.03     | 12  | 60  | 48    | 0.05 | -0.41    |
| MCQ – 30 Total | 67.78 | 15.96    | 30  | 117 | 87    | 0.31 | -0.29    |
| GAD – 7 Total  | 7.69  | 4.90     | 0   | 21  | 21    | 0.28 | -0.66    |

### *Interpretation*

The CSS–12 shows a highly symmetrical, nearly perfect bell-shaped distribution with a mean of 29.71 ( $SD = 9.03$ ). In contrast, the MCQ–30 displays the highest overall variability with a mean of 67.78 ( $SD = 15.96$ ) and a slight positive skew (0.31). The GAD–7 reflects a mild average anxiety level of 7.69 ( $SD = 4.90$ ). Across all three variables, the negative kurtosis values indicate platykurtic distributions.

**Table 2. Normality Testing – Shapiro-Wilk Test**

| Variable       | W Statistic | p-value |
|----------------|-------------|---------|
| CSS – 12 Total | 0.98        | 0.003   |
| MCQ – 30 Total | 0.98        | 0.01    |
| GAD – 7 Total  | 0.96        | <0.001  |

### ***Interpretation***

The assumption of normality was checked prior to analysis using the Shapiro-Wilk Test. The Shapiro-Wilk test indicated that CSS-12 ( $W = .98, p = .003$ ), MCQ-30 ( $W = .98, p = .01$ ), and GAD-7 ( $W = .96, p < .001$ ) significantly departed from normality.

**Table 3. Gender differences in CSS-12 scores using Wilcoxon Rank Sum Test**

| Variable       | W Statistic | p-value |
|----------------|-------------|---------|
| CSS – 12 Total | 6638.5      | 0.37    |

### ***Interpretation***

To test whether women had higher CSS-12 scores than men, a Wilcoxon rank-sum Test with continuity correction was conducted. The difference between the two groups was not statistically significant,  $W = 6638.5, p = .373$ .

**Table 4. Reliability Analysis of CSS-12, MCQ-30 and GAD-7 using Cronbach's alpha**

| Measure                         | No. of Items ( $k$ ) | Cronbah's alpha ( $\alpha$ ) | Interpretation        |
|---------------------------------|----------------------|------------------------------|-----------------------|
| Cyberchondria<br>Severity Scale | 12                   | .90                          | Very high reliability |
| Metacognitions<br>Questionnaire | 30                   | .92                          | Very high reliability |
| Generalized Anxiety<br>Disorder | 7                    | .88                          | High reliability      |



**Table 5. Spearman Rank Correlation between CSS-12, MCQ-30 and GAD-7**

| Variable | CSS – 12 | MCQ – 30 | GAD - 7 |
|----------|----------|----------|---------|
| CSS – 12 | -        |          |         |
| MCQ – 30 | .45***   | -        |         |
| GAD – 7  | .34***   | .63***   | -       |

Note. \*\*\*  $p < .001$

### ***Interpretation***

Spearman's rank-order correlations were computed to assess the relationship between cyberchondria (CSS-12), metacognitive beliefs (MCQ-30), and general anxiety (GAD-7). Significant positive correlations were found between the three variables ( $p < .001$ ). Strongest correlations were found between Metacognition and Generalized Anxiety,  $r_s(227) = .63$ ,  $p < .001$  whereas weak-to-moderate correlations were found between cyberchondria and metacognitive beliefs ( $r_s = .45$ ) and cyberchondria and general anxiety ( $r_s = .34$ ).

**Table 6. Multiple Linear Regression Predicting CSS – 12**

| <b>Variable</b> | <b><i>B</i></b> | <b>95 % CI</b> | <b><i>SE</i></b> | <b><math>\beta</math></b> | <b><i>t</i></b> | <b><i>p</i></b> |
|-----------------|-----------------|----------------|------------------|---------------------------|-----------------|-----------------|
| (intercept)     | 12.04           | [7.25, 16.83]  | 2.40             | -                         | 5.02            | < .001          |
| MCQ-30          | 0.25            | [0.16, 0.33]   | 0.05             | .44                       | 5.51            | < .001          |
| GAD-7           | 0.11            | [-0.17, 0.38]  | 0.15             | .06                       | 0.71            | .481            |

*Note.*  $N = 229$ ,  $R^2 = .23$ , Adjusted  $R^2 = .22$ , CI = Confidence Interval.

Standard Errors,  $t$ -statistics and  $p$ -values were calculated using HC3 Robust standard errors to account for heteroscedasticity.

### ***Interpretation***

The results revealed that the overall model was significant,  $F(2, 226) = 33.62$ ,  $p < .001$ . The model explained 22.9% of the variance in the CSS-12 scores ( $R^2 = .23$ , Adjusted  $R^2 = .22$ ). As seen in Table 4, only MCQ-30 significantly predicted CSS-12 ( $\beta = .44$ ,  $p < .001$ ) indicating that higher MCQ-30 scores were associated with higher CSS-12 scores. GAD-7 did not significantly contribute to the model ( $\beta = .06$ ,  $p = .481$ ).

Before interpreting the regression model, diagnostic analyses were conducted to verify that the assumptions of multiple linear regression were adequately satisfied.

**Table 7. Regression Diagnostics**

| <b>Assumption</b>                       | <b>Diagnostic Test</b>          | <b>Result</b>                            | <b>Interpretation</b>                                          |
|-----------------------------------------|---------------------------------|------------------------------------------|----------------------------------------------------------------|
| Linearity                               | Tukey Test                      | $p = .35$                                | Assumption met                                                 |
| Multicollinearity                       | Variance Inflation Factor (VIF) | VIF = 1.69 for both predictors (below 2) | No evidence of multicollinearity between MCQ – 30 and GAD – 7. |
| Influential Data Points                 | Cook's Distance                 | All $D_i < .08$ (below threshold of 1)   | No influential observations identified                         |
| Normality of Residuals                  | Q-Q Plot                        | Visual inspection                        | Approximate normal distribution of residuals                   |
| Homoscedasticity                        | Breusch-Pagan Test              | $\chi^2(2) = 7.23, p = .027$             | Assumption Violated                                            |
| Corrective Measure for Homoscedasticity | HC3 Robust Standard Errors      | Applied*                                 | Used to obtain robust coefficient estimates                    |

*Note.* \* **HC3 Robust Standard Errors** were utilized to ensure the reliability of the coefficients.

The results indicated that while MCQ-30 significantly predicted CSS-12 ( $B = .25, SE = .05, \beta = .44, t(226) = 5.51, p < .001$ ), GAD-7 did not contribute significantly ( $B = .11, SE = .15, \beta = .06, t(226) = .71, p = .481$ ).

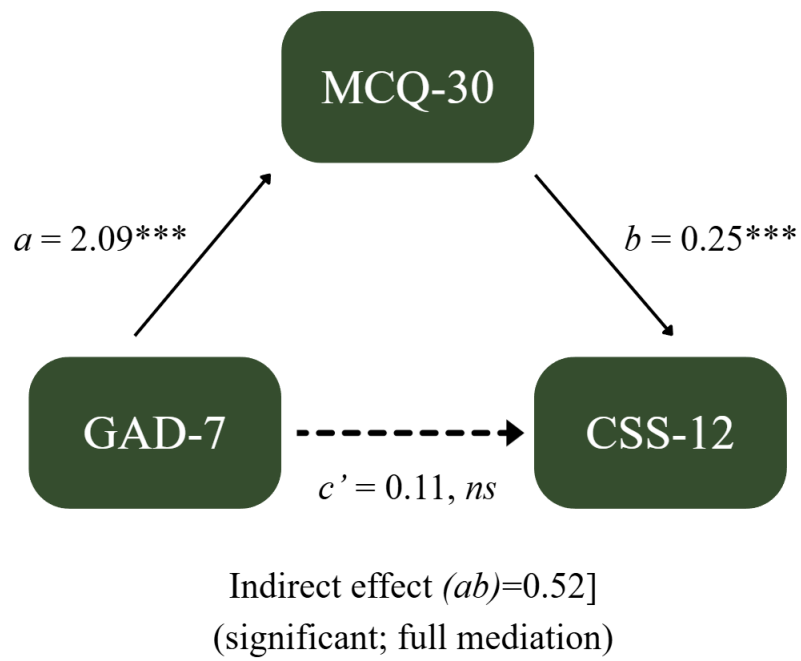
**Table 8. Mediation Model of GAD-7 on CSS-12 through MCQ-30**

| Predictor       | <i>B</i> | <i>SE</i> | <i>t</i> | <i>p</i> | 95% CI        |
|-----------------|----------|-----------|----------|----------|---------------|
| Path <i>a</i>   | 2.09     | 0.17      | 12.55    | < .001   | [1.76, 2.41]  |
| Path <i>b</i>   | 0.25     | 0.04      | 5.78     | < .001   | [0.16, 0.33]  |
| Path <i>c</i> ′ | 0.11     | 0.14      | 0.76     | .445     | [-0.17, 0.38] |
| Indirect Effect | Effect   | Boot SE   |          |          | Boot 95% CI   |
| Path <i>ab</i>  | 0.52     | 0.10      | -        | -        | [0.32, 0.73]  |

***Interpretation***

A mediation analysis was conducted using Hayes PROCESS macro (model 4) with 5,000 bootstrapped samples to check for the indirect effect of MCQ-30 (mediator variable) on the relationship between GAD-7 (independent variable) and CSS-12 (dependent variable) as seen in Table 5. GAD-7 significantly predicted the mediator MCQ-30,  $a = 2.09$ ,  $SE = 0.17$ ,  $t(227) = 12.55$ ,  $p < .001$ . The mediator MCQ-30 significantly predicted CSS-12 while controlling for GAD-7 ( $b = 0.25$ ,  $SE = 0.04$ ,  $t(226) = 5.78$ ,  $p < .001$ ) whereas GAD-7 did not significantly predict CSS-12 ( $c' = 0.11$ ,  $SE = 0.14$ ,  $t(226) = 0.76$ ,  $p = .445$ ). An indirect effect was tested using bootstrapped percentile confidence intervals. The indirect effect of GAD-7 on CSS-12 through MCQ -30 was found to be significant,  $ab = 0.52$ ,  $SE = 0.10$ , 95 % CI = [0.32, 0.73]. Since the confidence interval does not include zero, mediation is supported. Also, the direct effect  $c'$  was not significant whereas the indirect effect  $ab$  was significant, indicating full mediation.

**Figure 1. The Mediation Pathway**

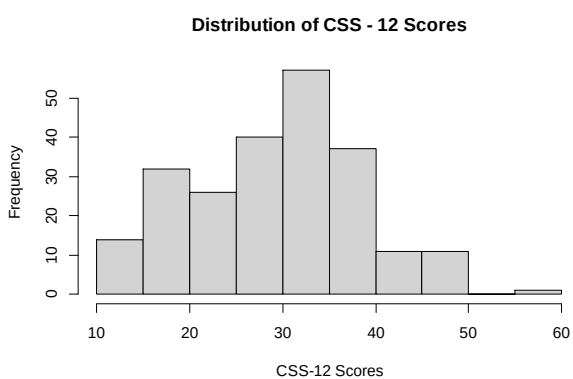


*Note.* \*\*\* $p < .001$ ; *ns* = not significant

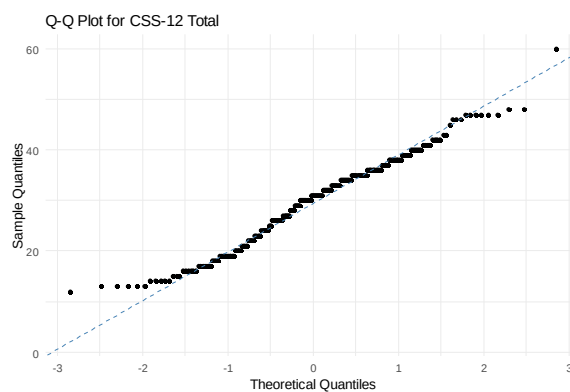
## VISUALIZATIONS

**Figure 2. Distribution and Normality of CSS-12, MCQ-30 and GAD-7 Scores**

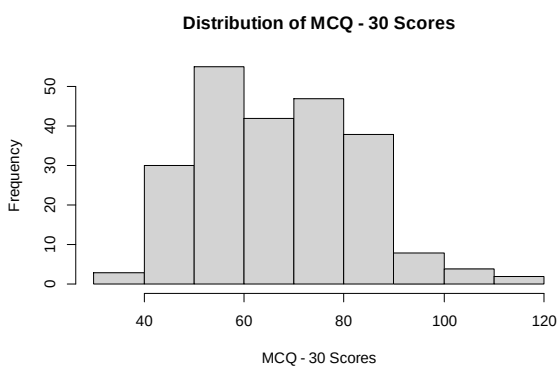
A



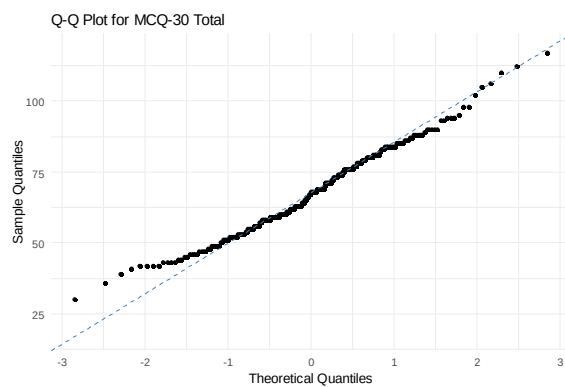
B



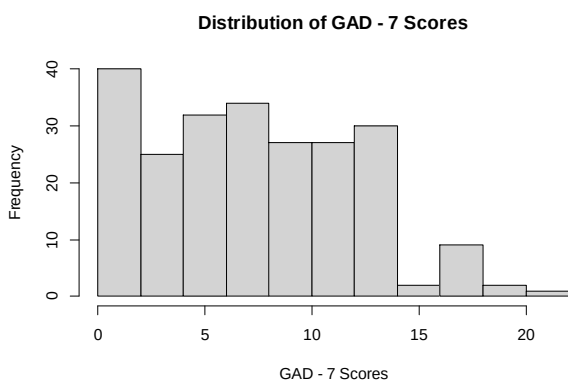
C



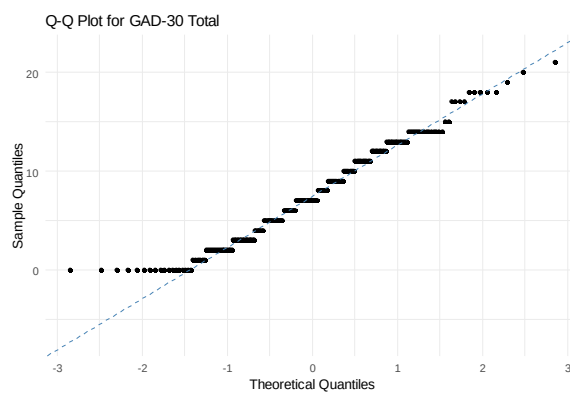
D



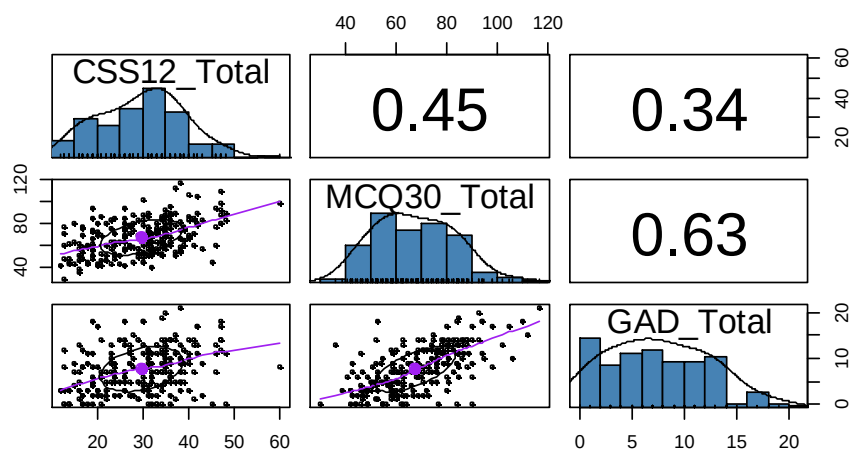
A



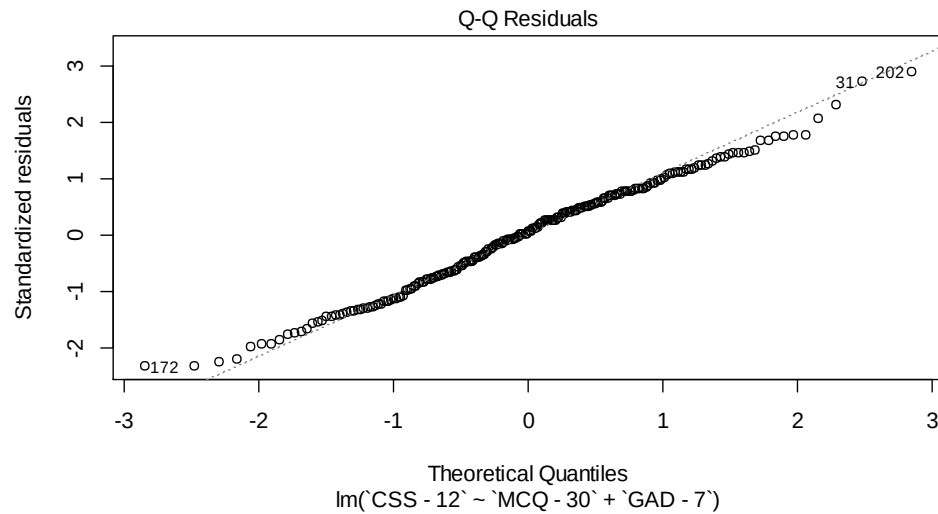
B



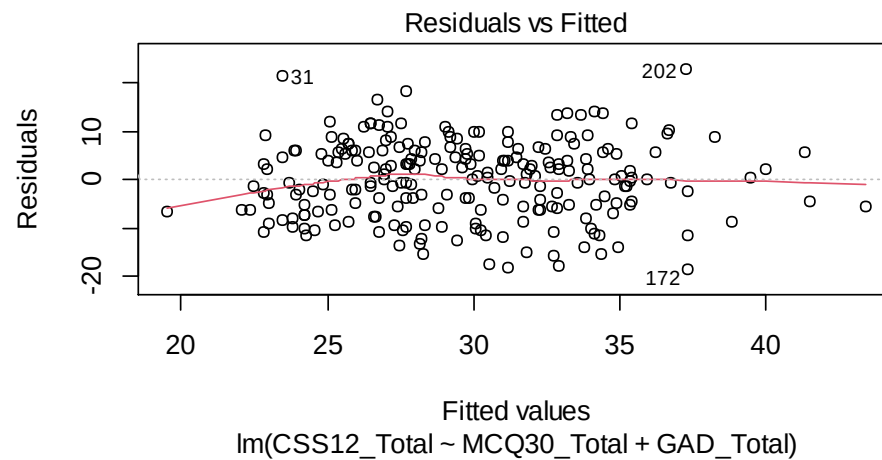
**Figure 3. Pairs Panel for CSS-12, MCQ-30 and GAD-7 Scores**



**Figure 4. Q-Q Plots for Checking Normal Distribution of Residuals**



**Figure 5. Residual vs Fitted Plot for Linearity and Homoscedasticity**





### KEY FINDINGS

- Metacognitive beliefs showed a stronger association with cyberchondria than general anxiety.
- In the regression model, metacognitive beliefs significantly predicted cyberchondria, whereas anxiety did not contribute significantly after controlling for metacognitive beliefs.
- Mediation analysis indicated that metacognitive beliefs explained the relationship between anxiety and cyberchondria.
- The findings suggest that dysfunctional metacognitive beliefs may represent an important mechanism underlying cyberchondria.

### TECHNICAL CAPABILITIES

| Competency           | Skills Demonstrated                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Collection      | Survey design and data collection using Google Forms                                                                                                       |
| Data Preparation     | Data cleaning and organization using Microsoft Excel                                                                                                       |
| Statistical Analysis | Reliability analysis, exploratory data analysis, normality testing, correlation, regression, mediation analysis and model diagnostics using R and RStudio. |
| Data Visualization   | Histograms, Q-Q plots, correlation matrix, regression diagnostic plots                                                                                     |

|                            |                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------|
| Research and Documentation | Statistical interpretation, report writing,<br>GitHub documentation and portfolio<br>development. |
| Software & Tools           | Google Forms, Google Sheets, Microsoft<br>Excel, R, RStudio and G*Power                           |